HR EMPLOYEE ATTRITION ANALYSIS

Power BI Project Report and Visual Insights

- **Submitted by: Srijai Nimmagadda**
TAH1708262282

1. INTRODUCTION

The HR Attrition Analysis dashboard provides an overview of employee attrition within an organisation, using the IBM HR Analytics Employee Attrition & Performance dataset. Attrition refers to employees leaving the organisation. High attrition increases recruitment and training costs and can affect productivity, which makes it an important measure for Human Resources teams.

The dashboard consists of two pages. The first page, **Overview**, presents headline KPIs and shows how attrition is distributed across departments, genders, age groups, and job roles. The second page, **Key Drivers**, examines factors that may be associated with attrition, including overtime, job satisfaction, work-life balance, business travel, monthly income, and years spent at the company.

Interactive Department and Gender slicers on both pages allow users to filter the visuals and examine attrition patterns for specific groups of employees.

The aim of the analysis is to identify which employee groups experience the highest attrition and which workplace factors show the strongest association with employees leaving.

2. DATA PREPARATION AND MODELLING

2.1 Data Preparation

The dataset was downloaded from Kaggle as a CSV file and imported into Power BI using **Get Data → Text/CSV**. It contains **1,470 records** (one per employee) and **35 columns**, covering demographics (age, gender, marital status), job details (department, job role, job level), compensation (monthly income, salary hike), work conditions (overtime, business travel), satisfaction scores, and tenure.

The data was reviewed in Power Query before any visuals were created:

- **Header check:** The first row was promoted to column headers, giving 1,470 data rows.
- **Data quality check:** Column quality and column profiling were enabled using the entire data set. All columns showed **100% valid values, 0% errors, and 0% empty values** (Figure 1).
- **Data types:** Text columns (Attrition, Department, JobRole, Gender, OverTime, BusinessTravel, etc.) and whole-number columns were checked and confirmed.

- **Removing constant columns:** Three columns contained the same value in every row and added no analytical value. These were removed:
    - EmployeeCount (always 1)
    - Over18 (always “Y”)
    - StandardHours (always 80)
- **Age Group column:** A custom column was added to group employees into five age bands: 18–25, 26–35, 36–45, 46–55, and 55+.

After these steps, the dataset contained **33 columns** and was loaded into the Power BI model.

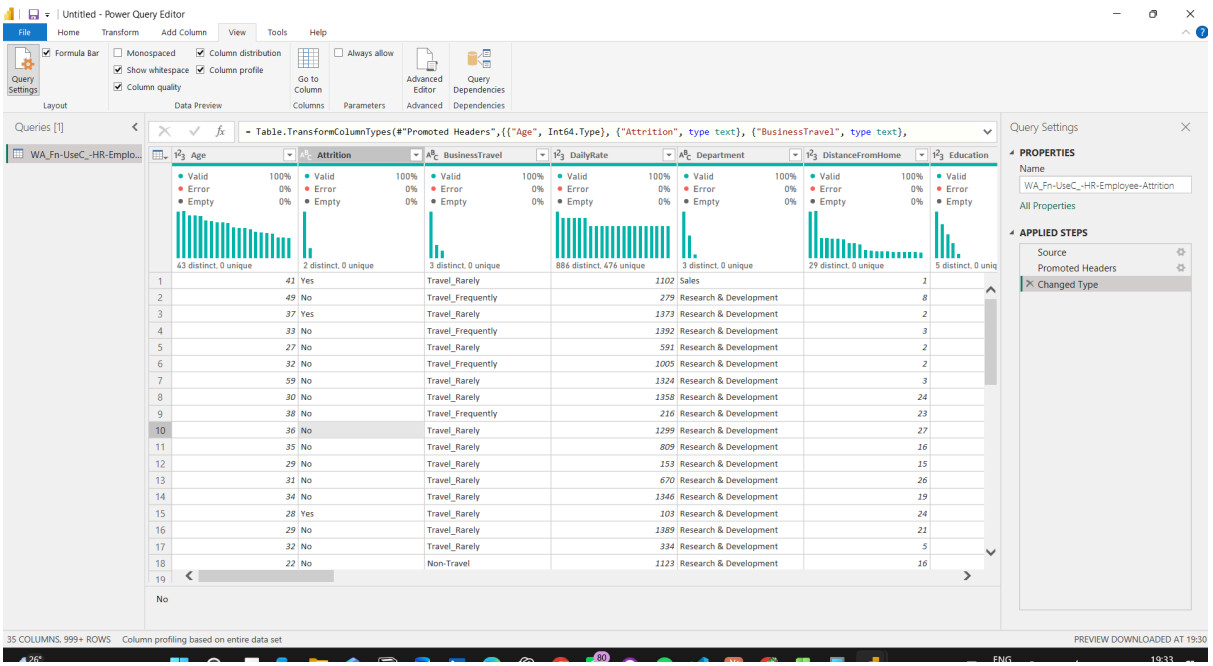


Figure 1: Data quality check in Power Query showing 100% valid values

2.2 Data Modelling

The table was renamed **HR** to keep formulas short and readable. Several columns in the dataset store ratings as numeric codes, which are difficult to interpret on charts. Four calculated columns were therefore created using the DAX SWITCH function to convert these codes into readable labels:

- **Education Level:** 1 = Below College, 2 = College, 3 = Bachelor, 4 = Master, 5 = Doctor
- **Job Satisfaction Level:** 1 = Low, 2 = Medium, 3 = High, 4 = Very High
- **Environment Satisfaction Level:** 1 = Low, 2 = Medium, 3 = High, 4 = Very High
- **Work Life Balance Level:** 1 = Bad, 2 = Good, 3 = Better, 4 = Best

Each label column was sorted by its original numeric column using **Sort by Column**, so charts display categories in logical order (for example, Low → Very High) rather than alphabetically.

The following DAX measures were created to calculate the KPIs:

Measure	DAX Formula
Total Employees	`COUNTROWS(HR)`
Attrition Count	`CALCULATE(COUNTROWS(HR), HR[Attrition] = "Yes")`
Active Employees	`[Total Employees] - [Attrition Count]`
Attrition Rate %	`DIVIDE([Attrition Count], [Total Employees], 0)`
Avg Age	`AVERAGE(HR[Age])`
Avg Monthly Income	`AVERAGE(HR[MonthlyIncome])`
Avg Years at Company	`AVERAGE(HR[YearsAtCompany])`

Age and income were calculated as **averages** rather than sums, because summing these fields would produce misleading totals. Attrition Rate % uses the DIVIDE function, which safely handles division by zero when a slicer selection returns no employees.

3. OVERALL DASHBOARD AND KPI INSIGHTS

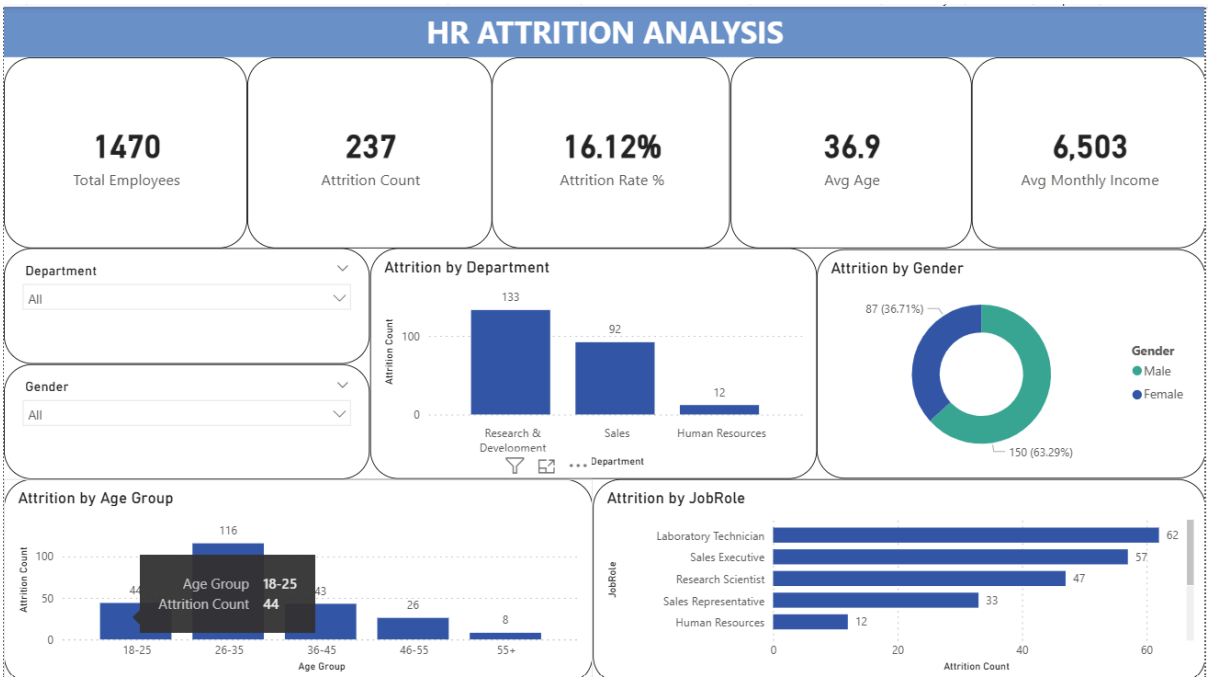


Figure 2: HR Attrition Analysis – Overview Dashboard (tooltip shown on the Age Group chart)

The Overview page combines five KPI cards with four visualisations and two slicers. The key performance indicators are:

- **Total Employees:** 1,470
- **Attrition Count:** 237
- **Attrition Rate:** 16.12%
- **Average Age:** 36.9 years
- **Average Monthly Income:** 6,503

These figures show that 237 of the 1,470 employees in the dataset have left the organisation, which means roughly **1 in every 6 employees**. The average employee is in their late thirties.

Figure 2 also shows the interactive **tooltip** feature. Hovering over the 18–25 bar of the Age Group chart displays the exact age group and its attrition count (44), so users can read precise values without relying only on data labels.

4. DETAILED INSIGHTS FROM THE VISUALS

4.1 Attrition by Department

The clustered column chart shows the number of employees who left from each department.

- **Research & Development** has the highest attrition count, with **133** employees leaving.
- **Sales** follows with **92**, and **Human Resources** has **12**.
- Counts alone can be misleading, because the departments differ greatly in size. Research & Development has 961 employees, Sales has 446, and Human Resources has 63.
- When department size is considered, **Sales has the highest attrition rate (about 20.6%)**, followed by Human Resources (about 19.0%) and Research & Development (about 13.8%).

This highlights why the Key Drivers page uses attrition **rates** rather than counts: a larger department will naturally show more leavers even if its employees are less likely to leave.

4.2 Attrition by Gender

The donut chart shows the gender split of employees who left.

- **Male** employees account for **150** leavers (**63.29%**).
- **Female** employees account for **87** leavers (**36.71%**).
- Male employees also make up a larger share of the workforce (882 of 1,470), so the gap in attrition **rates** is much smaller than the counts suggest: about **17.0%** for males compared with about **14.8%** for females.

Gender therefore appears to have only a modest association with attrition in this dataset.

4.3 Attrition by Age Group

The column chart shows attrition across the five age bands.

- The **26–35** age group has the highest attrition, with **116** leavers, which is nearly half of all attrition.
- The **18–25** group has **44** leavers, and **36–45** has **43**.
- Attrition falls steadily in older groups: **26** in the 46–55 group and **8** in the 55+ group.

Younger and early-career employees account for most of the attrition. The 18–25 group is relatively small in the workforce, so its 44 leavers may represent a high attrition rate. This could be explored further with a rate-based measure.

4.4 Attrition by Job Role

The bar chart ranks job roles by the number of employees who left.

- **Laboratory Technician** has the highest attrition count, with **62** leavers.
- **Sales Executive** (**57**) and **Research Scientist** (**47**) follow closely.
- **Sales Representative** (**33**) and **Human Resources** (**12**) have smaller counts, while senior roles such as Manager and Research Director have very few leavers.

Attrition is concentrated in **entry-level and mid-level operational roles** rather than senior positions. These roles are also typically lower-paid, which connects with the income insight in Section 5.5.

5. KEY DRIVERS OF ATTRITION

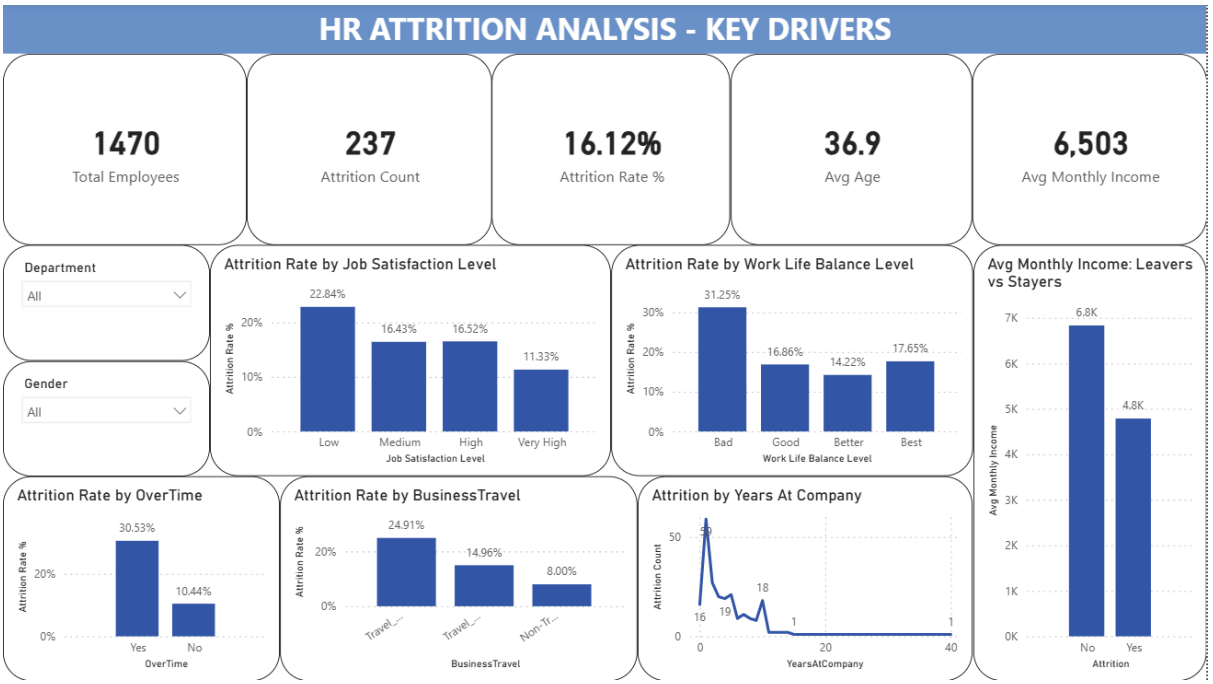


Figure 3: HR Attrition Analysis – Key Drivers Dashboard

The second page examines workplace factors that may be associated with attrition. All charts on this page, except the income and tenure charts, use **Attrition Rate %**, so that groups of different sizes can be compared fairly.

5.1 Attrition Rate by Overtime

- Employees who work overtime have an attrition rate of **30.53%**.
- Employees who do not work overtime have an attrition rate of **10.44%**.
- Employees working overtime are therefore about **three times more likely** to leave.

Overtime shows the **strongest association with attrition** of all the factors examined in this dashboard.

5.2 Attrition Rate by Job Satisfaction

- Employees with **Low** job satisfaction have the highest attrition rate at **22.84%**.
- **Medium** (16.43%) and **High** (16.52%) satisfaction show similar rates.
- **Very High** satisfaction has the lowest rate at **11.33%**.

Attrition generally decreases as job satisfaction increases. Employees with low satisfaction leave at about twice the rate of those with very high satisfaction.

5.3 Attrition Rate by Work-Life Balance

- Employees rating their work-life balance as **Bad** have the highest attrition rate at **31.25%**.
- The rate drops sharply for **Good** (16.86%) and **Better** (14.22%).
- **Best** shows a slight rise to **17.65%**.

Poor work-life balance is strongly associated with attrition. However, the “Bad” category contains only 80 employees, so this rate is based on a small group and should be interpreted with some caution. The slight rise for “Best” suggests that the relationship is not perfectly linear.

5.4 Attrition Rate by Business Travel

- Employees who **travel frequently** have an attrition rate of **24.91%**.
- Employees who **travel rarely** have a rate of **14.96%**.
- **Non-travelling** employees have the lowest rate at **8.00%**.

Attrition rises clearly with travel frequency. Frequent travellers leave at roughly **three times** the rate of employees who do not travel.

5.5 Average Monthly Income: Leavers vs Stayers

- Employees who **stayed** earn an average of about **6.8K** per month.
- Employees who **left** earn an average of about **4.8K** per month.
- Leavers earn roughly **2,000 less per month**, which is around **30% lower** income.

Lower pay is associated with higher attrition. This is consistent with Section 4.4, where most attrition occurred in entry-level roles.

5.6 Attrition by Years at Company

- Attrition is highest among employees with **1 year** at the company, with **59** leavers, which is about a quarter of all attrition.
- Employees with less than one year of service account for a further **16** leavers.
- A smaller secondary peak appears at **10 years** (**18** leavers).
- Beyond 10 years, attrition is very low, with only isolated cases up to 40 years.

Most employees who leave do so **early in their tenure**. This suggests that the first one to two years are the most critical period for retention.

6. INTERACTIVE FILTERS AND DASHBOARD EXPLORATION

Both dashboard pages include two dropdown slicers.

Department Slicer: Filters all KPI cards and charts on the page to a selected department, such as Sales, Research & Development, or Human Resources. For example, selecting Sales shows how overtime, satisfaction, and income patterns apply specifically to the sales team.

Gender Slicer: Filters the page to male or female employees, allowing comparison of attrition patterns between genders.

Together with the tooltips (shown in Figure 2), these filters allow users to explore different segments of the workforce without modifying the underlying data. All slicers were cleared before the dashboard screenshots were taken, so the figures in this report reflect the complete dataset.

7. CONCLUSION

The HR Attrition Analysis project uses Power BI to turn raw employee data into clear, interactive insights. Of 1,470 employees, **237 have left, giving an overall attrition rate of 16.12%**.

The Overview page shows that attrition is concentrated among **younger employees (26–35)**, in **entry-level roles** such as Laboratory Technician and Sales Executive, and in the **Sales department**, which has the highest attrition rate despite having fewer leavers than Research & Development.

The Key Drivers page identifies several factors strongly associated with attrition:

- **Overtime:** 30.53% attrition compared with 10.44% without overtime
- **Poor work-life balance:** 31.25% attrition for “Bad”
- **Frequent business travel:** 24.91% compared with 8.00% for non-travellers

- **Low job satisfaction:** 22.84% compared with 11.33% for “Very High”
- **Lower income:** leavers earn about 30% less on average
- **Short tenure:** the largest number of leavers have only 1 year at the company

These findings show **associations, not proof of cause**. Several factors may overlap; for example, junior employees may also earn less and work more overtime.

Recommendations based on the analysis:

1. **Monitor and limit overtime**, especially in high-attrition roles.
2. **Strengthen onboarding and early-career support** during the first two years of employment.
3. **Review pay** for entry-level roles such as Laboratory Technician and Sales Representative.
4. **Track job satisfaction and work-life balance** through regular surveys, and act early on low scores.

Overall, this project demonstrates how data preparation, DAX measures, and interactive visualisations in Power BI can help HR teams understand why employees leave and where to focus retention efforts.